Muhammad Rifqy

GIS Specialist | Remote Sensing & Land Cover Analysis | Forestry MRV & Carbon Project Spatial Data

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SUMMARY

GIS specialist with 3+ years of end-to-end spatial data production in large-scale forestry and coastal carbon programs. Experienced in designing remote sensing pipelines for land cover classification, habitat change detection, and project boundary delineation — with outputs supporting MRV reporting under World Bank and KEMENHUT frameworks. Proficient in multi-source satellite imagery analysis (Sentinel-1 SAR, Sentinel-2), UAV photogrammetry (Agisoft Metashape), and geospatial programming (Python: GeoPandas, Rasterio, GDAL; Google Earth Engine). Demonstrated record of delivering spatially accurate, audit-ready datasets with systematic QC and accuracy assessment workflows. Passionate about applying geospatial technology to environmental sustainability and climate mitigation.

SKILLS & TECHNICAL COMPETENCIES

UAV / Photogrammetry	Agisoft Metashape (orthomosaic, DEM, DSM, point cloud), Dronelink (flight planning)
Satellite Imagery	Sentinel-1 SAR, Sentinel-2 multispectral, Landsat; land cover classification, change detection, spectral index analysis (NDVI, NDWI, EVI, MVI)
GIS Platforms	ArcGIS Pro/Desktop, ArcGIS Online, Survey123, QGIS, Global Mapper
Geospatial Programming	Python (GeoPandas, Rasterio, GDAL, scikit-learn, Folium), R (sf, terra, ggplot2, rayshader), Google Earth Engine (GEE)
Web Mapping	Kepler.gl, Mapbox Studio, QGIStoWeb, ArcGIS Online, OpenStreetMap
Data & QC	Shapefile/GeoJSON database management, accuracy assessment, error matrix, QC/validation workflows, Survey123 field data pipelines
Visualization	Thematic maps, infographics, dashboards, cartographic production to Ministry P.6/PKTL standards
Languages	Indonesian (Native), English (Professional)

EXPERIENCE

Forestry Technical Staff for MRV (Contract) · Indonesia's FOLU NET SINK 2030 *June 2026 – Present*

- Supporting the development of MRV methodology documentation and structural guidance for Indonesia's FOLU Net Sink 2030 program, contributing to forestry sector mitigation reporting frameworks.
- Assisting in compilation and systematic documentation of technical data related to forestry sector activities and GHG mitigation contributions in alignment with national FOLU Net Sink 2030 targets.

Technical Mangrove Rehabilitation Facilitator (Contract) · M4CR Project — KEMENHUT / World Bank *May 2024 – Apr 2026*

- Designed and maintained project boundary spatial datasets for 5,800+ ha of mangrove rehabilitation sites across 5 regencies in Riau, with all outputs compliant with KLHK/KEMENHUT mapping standards (P.6/PKTL) — directly analogous to carbon project area delineation requirements.
- Developed multi-source remote sensing workflows integrating Sentinel-1 SAR and Sentinel-2 spectral indices (NDWI, EVI, MVI) for land cover classification, habitat change detection, and mangrove suitability mapping — supporting spatial evidence for World Bank PBC disbursement reporting.
- Performed systematic accuracy assessment and validation (error matrices) on all geospatial datasets, ensuring spatial integrity and attribute completeness across 100+ community group datasets for program accountability.
- Processed UAV imagery into DEM, orthomosaic, and DSM outputs using Agisoft Metashape; engineered YOLOv8-based instance segmentation pipeline achieving 90.4% precision for invasive species (Acrostichum aureum) mapping across 1,400 ha.
- Built and managed centralized Shapefile Database System serving as primary GIS infrastructure for PPIU M4CR Riau, including spatial data governance, QC workflows, and field data pipelines via Survey123.
- Produced high-quality thematic maps, infographics, and spatial visualizations to communicate findings to KLHK/KEMENHUT, World Bank stakeholders, and NGO partners in English and Bahasa Indonesia.
- Co-authored biophysical M&E reports covering 56 community groups across 1,683 ha, applying multi-method field data collection (census, systematic sampling, interviews) to assess plant survival rates for World Bank PBC disbursement reporting.

- Conducted drone-based monitoring and visual verification across 18 villages in Indragiri Hilir, producing integrated spatial-visual evidence packages for program accountability and donor reporting.
- Conducted advanced geospatial analysis integrating multi-source environmental data (hydrological connectivity, soil composition, flood inundation depth) to support ecological assessment and habitat restoration planning under IFC Environmental and Social Performance Standards.

Staff Teknis Pengolahan Data Spasial (GIS/Mapping) · CV. Pardus Arogya Samasta *May 2023 – Oct 2023*

- Produced watershed (DAS) maps following national cartographic standards; collected, analyzed, and interpreted geographic data for final map, graphics, and report outputs.
- Applied remote sensing techniques for watershed boundary mapping and land cover analysis.

Student Intern — Land Use Analyst · Pusat Pengendalian Pembangunan Ekoregion Jawa (P3EJ) *Mar 2020 – May 2020*

- Digitized 5,000+ land-use polygons at sub-1:5,000 scale across Bogor and Cirebon regencies, integrating land ownership, plantation, and topographic layers for provincial environmental carrying-capacity assessments.
- Ensured spatial accuracy, attribute completeness, and data consistency across all outputs in accordance with national mapping standards.

Research Assistant — Environmental Geomorphology Lab · Universitas Gadjah Mada *2021 – 2024*

- Developed landslide analysis techniques using Multinomial Logistic Regression–Regression Kriging and prepared geocomputational-geostatistics training modules (R programming) for geomorphology practicum courses.

Graphic & Data Visualizer (Volunteer) · Extinction Rebellion Riau Chapter · Aug 2023 – Jul 2025

EDUCATION

Universitas Riau *Aug 2025 – Present*

M.M.B. — Disaster Management (Peatland & Coastal Specialization) | GPA 4.00/4.00

Relevant courses: Mangrove Conservation, Flood and Abrasion, Forest & Peatland Fires, Peatland Ecological Disaster Studies, Disaster Management Information System & GIS

Universitas Gadjah Mada *Aug 2017 – Nov 2022*

S.Si. — Environmental Geography (B.Sc. equivalent) | GPA 3.49/4.00

Relevant courses: Geomorphological Survey Mapping, Environmental Geomorphology, Ecology & Environmental Study, Tropical Landscape Ecology, Environmental Management, GIS

COURSES & TRAINING

Year	Course	Provider	Type
2025	Bootcamp WebGIS Development	MAPID	Bootcamp
2024	QGIS: Pemrosesan dan Aplikasi Citra Satelit Resolusi Tinggi	Geoaccess Indonesia	Bootcamp
2024	GEE Tematik: Pemetaan Lahan Mangrove dan Carbon Stock	Geoaccess Indonesia	Course
2024	GEE Tematik: Monitoring Perubahan Garis Pantai dan Prediksinya	Geoaccess Indonesia	Course
2022	Cartography	ESRI MOOC	MOOC
2022	Going Places with Spatial Analysis	ESRI MOOC	MOOC
2021	R Fundamental for Data Science	DQLab	Course
2021	Python Fundamental for Data Science	DQLab	Course

PROJECTS & PUBLICATIONS

Independent Cartographic & Geospatial Research (med/rif maps) · *Oct 2023 – Present*

- Experiments with advanced cartographic techniques using R (rayshader, ggplot2), Python, and GEE; portfolio at instagram.com/medrifmaps and mamedrifqy.github.io.

GeoLab Notebook Repository · Python, GEE & R notebooks for geospatial analysis and remote sensing — mamedrifqy.github.io/geolab

Publication: Pemanfaatan Inovasi Geospasial Berbasis Sentinel-2 dan UAV untuk Optimalisasi Rehabilitasi Mangrove dalam Kerangka Program M4CR di Provinsi Riau. ResearchGate Preprint, Jan 2025. DOI: 10.13140/RG.2.2.21992.53762

Publication: Assessment of Ecosystem Services and Environmental Degradation by Using Landscape Approach in The Riparian Area of Lake Rawapening. Jurnal GEA, Oct 2022. DOI: 10.17509/gea.v22i1.39976

HONORS & AWARDS

2022 · Best Map, Spatialnode Map of The Week (Global)

2017–2021 · Scholarship Awardee, Governance of Riau Province (Penelusuran Bibit Unggul Kemitraan Program)